

RUBE GOLDBERG MACHINES

Design, Build, and Invent Your Own Chain Reaction!

WHAT YOU WILL LEARN

- **Understand Chain Reactions:** Learn how one moving object triggers another in a planned sequence.
- **Apply Cause & Effect:** Map out exactly how pushing one item causes the next event to happen.
- **Sequence Events:** Master the vocabulary of sequential design using logical transition words.
- **Embrace Engineering:** Plan, build, test, and confidently adjust your machine through trial and error.

MATERIALS CHECKLIST

Gather these everyday objects around you to build your engineering setup:

- 2-3 Hardcover books (great for building ramps)
- 1 Small ball or marble (your main moving starter tool)
- 3-4 Blocks, steel tumblers, or food packets (to line up like dominoes)
- 1 Finale object (a small bell/ghanti, a miniature toy car, or a mini flag)
- Rulers or pencils (perfect for building levers or barriers)
- Masking tape, scissors, or sticky notes

THE WONDER CHALLENGE

 **Wonder Question:** Can you ring a bell or move a final object **WITHOUT** using your hands directly?

Your goal today is to engineer a mechanism that starts with a single movement and travels all the way to cross the finish line to complete a simple task automatically!

CORE CONCEPTS

What is a Rube Goldberg Machine?

It is a crazy, fun, and intentionally complex machine that solves a very simple task in a long, roundabout chain reaction!

Cause & Effect Relationship:

Every single step in your machine relies on cause and effect. The **Cause** is the action that makes something happen (e.g., a ball hitting a block). The **Effect** is the result that follows (e.g., the block tipping over).

THE ENGINEERING BLUEPRINT: 4-STEP SEQUENCE

To keep your chain reaction organized and clear, describe each action using your core tracking words:



Example Story Blueprint (Arjun's Biscuit Story):

1. **FIRST** Arjun accidentally knocks his school bag over.
2. **THEN** The tumbling bag hits a water bottle standing nearby.
3. **NEXT** The rolling water bottle pushes a box of biscuits straight off the desk ledge.
4. **FINALLY** The biscuit packet lands safely right in front of a hungry puppy waiting below!

YOUR BUILDING PROCESS

Follow these steps to bring your machine to life:

- **Step 1: Sketch the Concept:** Draw your plan on a quick note paper. Label at least 3 distinct moving steps using your sequencing words. Your layout must start with a ball or marble roll and finish with your chosen finale task.
- **Step 2: Construct & Adjust:** Assemble your items! Pay careful attention to structural spacing—placing your objects too far apart is the number one reason chain reactions stop short.

- **Step 3: Test & Debug:** Run trial runs. If something fails to fall over, don't worry! Real-world aerospace and mechanical engineers constantly encounter failed attempts. Analyze which link in the chain broken, adjust your angles, and try again.

VOCABULARY CARD

Key Word	Meaning & Description
Chain Reaction	A series of events where each separate action triggers the next consecutive one to occur.
Cause	The intentional action or movement that directly makes a subsequent thing happen.
Effect	The result or consequence that directly follows an initial cause.
Sequence	The clean timeline order in which events occur: First → Then → Next → Finally.
Engineer	A creative problem-solver who designs, builds, evaluates, and continuously improves physical inventions.

HOME DETECTIVE CHALLENGE

Tonight's Exploration:

Search around your house for a real-life chain reaction (either an accidental spill or a purposeful mechanism). In your personal science notebook, sketch the event and map out exactly how it traveled:

FIRST → What action initiated the movement?

THEN → What item was directly struck next?

NEXT → What followed as a direct result?

FINALLY → What was the closing action or end result?